Week 3 Lecture 6

Probability

Probability Theory and Mathematical Statistics

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Sample Space

A collection of all elementary results, or outcomes of an experiment, is called a sample space, which we usually write Ω.

1. ✏️ We have three playing cards: **queen**, **king**, and **jack**. All are shown face down, and one is chosen at random and turned up. What is the set of outcomes?

   → $\Omega = \{Q, K, J\}$

2. ✏️ We choose two cards replacing the card we have chosen. What is the sample space?

   → $\Omega = \{QQ, QK, QN, KQ, KK, KJ, JQ, JK, JJ\}$

🖊 A Choice of Strategy for Planning a Family

A couple decides to have children. As they know no mathematics, they decide to have children until a girl then a boy are born.

What is the sample space?

Does this strategy bound the number of children they could be planning to have?

Event

Any set of outcomes is an event. Thus, events are subsets of the sample space.

$$E \subseteq \Omega$$

1. ✏️ What is an event of not taking king from $\Omega = \{Q, K, J\}$?

   → $E = \{Q, J\}$

2. ✏️ What is an event of taking queen at least ones from two takes?

   → $\{QQ, QK, QN, KQ, JQ\}$

Formal definition of Probability

σ-Algebra

A collection $\mathfrak{S}$ of events is a sigma-algebra on sample space Ω if:

(a) It includes the sample space: $\Omega \in \mathfrak{S}$.

(b) Every event in M is contained along with its complement; that is, $E \in \mathfrak{S} \Rightarrow E^c \in \mathfrak{S}$.

(c) Every finite or countable collection of events in M is contained along with its union; that is, $E_1, E_2, \ldots \in \mathfrak{S} \Rightarrow \bigcup_i E_i \in \mathfrak{S}$.

Degenerate σ-Algebra and Power set

By conditions (a) and (b) every σ-algebra has to contain the sample space Ω and the empty event $\varnothing$.

This minimal collection $\mathfrak{S} = \{\Omega, \varnothing\}$ forms a sigma-algebra that is called degenerate.

The richest σ-algebra on a sample space Ω is the collection of all the events:

$$\mathfrak{S} = 2^{\Omega} = \{E \,|\, E \subseteq \Omega\}.$$

Probability

Assume a sample space Ω and a sigma-algebra of events $\mathfrak{S}$ on it.

Probability

$$P : \mathfrak{S} \to [0, 1]$$

is a function of events with the domain $\mathfrak{S}$ and the range $[0, 1]$ that satisfies the following two conditions:

(Unit measure) The sample space has unit probability: $P(\Omega) = 1$.

(Sigma-additivity) For any finite or countable collection of mutually exclusive events $E_1, E_2, \ldots \in \mathfrak{S}$,

$$P(E_1 \cup E_2 \cup \ldots) = P(E_1) + P(E_2) + \ldots$$

Extreme Cases of Probability

A sample space Ω consists of all possible outcomes, therefore, it occurs for sure. On the contrary, an empty event $\varnothing$ never occurs. So,

$$P(\Omega) = 1 \quad \text{and} \quad P(\varnothing) = 0.$$

Equally Likely Outcomes

Let the sample space Ω consists of n possible outcomes, $\omega_1, \ldots, \omega_n$, each having the same probability. Since

$$\sum_{k=1}^{n} P(\omega_k) = P(\Omega) = 1,$$

we have in this case $P(\omega_k) = 1/n$ for all k.

Further, the probability of any event E equals:

$$P(E) = \frac{\dim E}{\dim \Omega} = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}} = \frac{N_F}{N_T},$$

where index "F" means "favorable" and "T" means "total."

Tasks

1. A card is drawn from a bridge 52-card deck at random. Compute the probability that the selected card is a spade.
2. A young family plans to have two children. What is the probability of two girls?
3. A box contains 500 envelopes, of which 75 contain €100 in cash, 150 contain €25, and 275 contain €10. An envelope may be purchased for €2. Find the probability that the first envelope will give profit more than €10.

Sampling with/without replacement

Sampling **with replacement** means that every sampled item is replaced into the initial set, so that any of the objects can be selected with probability $1/n$ at any time. In particular, the same object may be sampled more than once.

Sampling **without replacement** means that every sampled item is removed from further sampling, so the set of possibilities reduces by 1 after each selection.

Distinguishable and indistinguishable

Objects are **distinguishable** if sampling of exactly the same objects in a **different order yields a different outcome**, that is, a different element of the sample space.

For **indistinguishable** objects, the order is not important, it only matters which objects are sampled and which ones aren't.
Indistinguishable objects arranged in a **different order do not generate a new outcome**.

Permutations with Replacement

Possible selections of k distinguishable objects from a set of n are called permutations. When we sample with replacement, each time there are n possible selections, hence the total number of permutations is:

$$P_r(n, k) = \underbrace{n \cdot n \cdot \ldots \cdot n}_{k \text{ terms}} = n^k$$

✏️ If an 8-character password consists of digits and letters, how much longer would it take to brute force the password if special characters were added to it?

Permutations without Replacement

During sampling without replacement, the number of possible selections reduces by 1 each time an object is sampled.

Therefore, the number of permutations is:

$$P(n,k) = \underbrace{n(n-1)(n-2)\cdot \ldots \cdot (n-k+1)}_{k \text{ terms}} = \frac{n!}{(n-k)!}$$

✏️ In how many ways can 10 students be placed in queue of 4 people in Muna?

Combinations without Replacement

Possible selections of k indistinguishable objects from a set of n are called combinations.

The only difference from $P(n, k)$ is disregarding the order.

$$C(n, k) = \binom{n}{k} = \frac{P(n, k)}{P(k, k)} = \frac{n!}{k!(n-k)!}$$

✏️ An antivirus software reports that 3 folders out of 10 are infected.

How many possibilities are there?

Combinations with Replacement

Combinations with replacement are denoted as $C_r(n, k)$ and represented by $\binom{n+k-1}{k}$.

$$C_r(n, k) = C(n + k - 1, k) = \binom{n + k - 1}{k} = \frac{(n + k - 1)!}{k!(n - 1)!}$$